Transfer: circuits and induction

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Apply circuit relationships and explain an induction observation.

SUCCESS CRITERIA - I can...

1. Use current and voltage rules in a new circuit.
2. Account for electrical power and energy.
3. Explain induction through changing flux.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “resistance” and “power” to explain your answer. Sentence starters are optional.

resistance: voltage divided by current for a stated operating condition

power: rate of energy transfer

induction: producing an emf through a changing magnetic flux

01 Warm-up: bring back an earlier idea

A device receives **50 J** mechanical energy at its shaft and produces electrical energy at its terminals. **Name it** and **name one possible loss**.

Sentence starter: I notice … . This supports … because … .

Marzano check - circle: ① ② ③

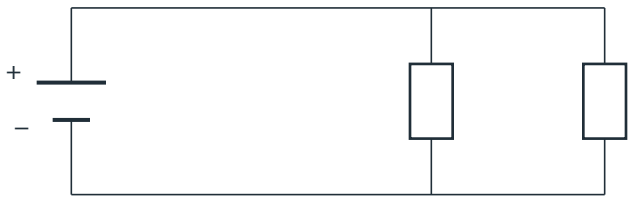
02 Worked example: explain the prediction

Reference: $V = IR$; $P = IV$; $E = P\Delta t$; $|\text{emf}_{avg}| = N|\Delta\Phi|/\Delta t$. Use stated facts about the circuit or coil. Ideal battery and wires unless stated otherwise.

Worked reasoning: A **12 V** supply drives a **6 ohm** resistor: current **2 A** and power **24 W**. A complete explanation checks both the charge-flow rate and the energy-transfer rate.

Your turn: 1. **Draw the circuit** and an energy-flow arrow.
2. **Label what an ammeter** and voltmeter would measure and where each connects.

Sentence starter: In my work, I used … to show … . I labeled it … .



Label component values, current and voltage as needed.

Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

Sketch two resistors in parallel across a battery and **label the equal-voltage branches**.

Sentence starter: In my work, I used … to show … . I labeled it … .



Circuit symbols: long/short lines = battery; rectangles = resistors.

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

04 Apply the physics

A **6 V** battery powers **3 Ω** and **6 Ω** resistors in parallel.
1. **Label both branch voltages**.
2. **Find each current**, total source current and total power $P = VI_{total}$.

Sentence starter: I found … [unit]. This value means … in this situation.



Label component values, current and voltage as needed.

Marzano check - circle: 1 2 3

05 Explain your reasoning

A magnet is held motionless near a fixed coil, then moved into it. **Explain the change** in the meter response without saying the magnet supplies unlimited energy.

Sentence starter: I think … . I can point to …, which shows …, so … .

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A **4 Ω** resistor carries **6 A**. **Find the potential difference across** it.
Draw the stated circuit and **label quantities relevant** to the question.

Sentence starter: Both … . They differ in … because … .

Draw the complete circuit and label shared nodes.

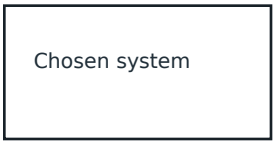
Marzano check - circle: 1 2 3

07 Exit check

A motor uses **6 W** for **5 s**. **Find energy** and **name one possible non-mechanical output**.

Sentence starter: I found … [unit]. This value means … in this situation.

Draw and label all energy transfers across the boundary.



Stored energy change: _____ J

Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

Replace the resistor with two **12 ohm** resistors in parallel. **Compare total current** and power, then **draw both branch currents**.

Sentence starter: Both … . They differ in … because … .



Label component values, current and voltage as needed.

Marzano check - circle: 1 2 3